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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

V Semester B. E. Regular Examinations Feb/Mar -2025

Common to CSE/ISE/CD/CY

THEORY OF COMPUTATION

Time: 03 Hours *Maximum Marks: 100*

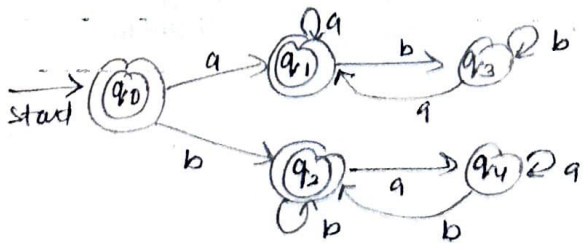
Instructions to candidates:

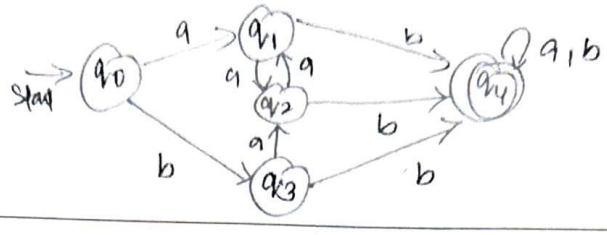
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Design a DFA to accept language $L = \{w \mid w \bmod 3 < 2\}$.	02	3	3
	1.2	What are distinguishable and equivalent states?	02	1	1
	1.3	What is the language of the regular expression means $(0^*1 + 1^*0)^*0$?	02	2	1
	1.4	Define parse tree. Construct parse tree for the string $aaabaabba$ for the grammar with productions. $S \to SS \mid bTT \mid TbT \mid TTb \mid \in$ $T \to aS \mid SaS \mid Sa \mid a$	02	2	2
	1.5	If L is CFL & R is regular then prove that $L \cap R$ is CFL.	02	1	2
	1.6	Give the transition table for DPDA recognizing the following language $L = \{a^n b^{n+m} a^m \mid n, m \geq 1\}$	02	2	2
	1.7	What is the equivalent left linear grammar for the following given right linear grammar $S \to abA \mid bB \mid aba, A \to b \mid aB \mid bA, B \to aB \mid aA$	02	3	3
	1.8	Differentiate recursively enumerable language and recursive languages.	02	1	1
	1.9	Consider homomorphism h from alphabet $\{0,1,2\}$ to $\{a, b\}$ defined by $G(0) = ab, h(1) = b$ and $G(2) = aa$. Find $h(0210)$ and $h^{-1}(ababb)$	02	2	2
	1.10	Construct a TM to double a binary integer.	02	4	3

PART-B

2	a	Convert the following DFA to regular expression. 	06	3	2
	b	Define DFA, language accepted by DFA and construct a minimal DFA which accepts set of all strings over $\{0,1\}$ which when interpreted as a binary number is divisible by 4.	05	4	3

	c	Minimize the states of the following DFA. 	05	3	3
3	a	State and prove pumping lemma for regular language: show that the following language is not regular $L = \{0^n \mid n \text{ is prime}\}$.	06	2	2
	b	Define CFG and write a CFG for the language $L = \{a^i b^j c^k \mid j = i \text{ or } j = k\}$	05	3	4
	c	Convert the below grammar to CNF $S \to ASB \mid \epsilon, A \to aAS \mid a, B \to SbS \mid A \mid bb$	05	2	3
		OR			
4	a	Show that the regular languages are closed under union and complement operation.	06	2	2
	b	Define regular grammar and obtain right linear grammar for the language $L = \{a^n b^n \mid n \geq 2, m \geq 3\}$	05	2	3
	c	Define GNF. Convert the given grammar $S \to AB \mid BC, A \to aB \mid bA \mid a, B \to bB \mid cC \mid b\ C \to C$ into GNF.	05	2	2
5	a	Let L be $L(M_1)$ for some PDA with empty stack $M_1(Q_1, \Sigma, \Gamma, \delta, q_0, z_0, F)$, prove that there exist a final state PDA M_2 such that $L = L(M_2)$. Construct a PDA for $L = \{a^n b^n \mid n \geq 1\} \cup \{a^n b^{2n} \mid n \geq 1\}$	06	3	3
	b	State and prove pumping lemma for CFL.	05	2	2
	c	For the grammar $S \to 0S1 \mid A, A \to 1A0 \mid S \mid \epsilon$ obtain the corresponding PDA. Show that the string 0101 is accepted by the PDA and it is generated by the equivalent CFG.	05	2	2
		OR			
6	a	What are the steps to be steps to be followed while finding on equivalent CFG from the given PDA by empty stack? Obtain a CFG that generates the language accepted by PDA $M = (\{q_0, q_1\}, \{a, b\}, \{A, Z\}, \delta, q_0, Z, \{q_1\})$ with the transitions $\delta(q_0, a, z) = \{(q_0, AZ)\}$ $\delta(q_0, b, A) = \{(q_0, AA)\}$ $\delta(q_0, a, A) = \{(q_1, \epsilon)\}$	06	2	3
	b	Let $L_1 = \{a^i b^j c^k \mid i < j\}$ & $L_2 = \{a^i b^j c^k \mid i < k\}$ show that L_1 and L_2 are context free but $L_3 = L_1 - L_2$ is not context free.	05	3	3
	c	Define $DPDA$. Construct $DPDA$ for $L = \{WCW^R \mid w \in \{a, b\}*\}$. Show by IDs the string *abacaba* is accepted.	05	3	3
7	a	Define Turing machine and language of TM. Design a TM such that $(q0, BwB) \vdash* (qf, Bw\,Bw^r\,B)$ where $q0$ is the initial state, qf is final state, B is blank and $W \in \{0,1\}^*$.	08	4	4
	b	Write short note on the following: i) Multi-stack machine ii) Counter machine.	08	1	1
		OR			

8	a	Discuss Multi tape Turing Machine and multi track Turing machine in detail.	08	1	1
	b	Let x & y are two positive integers represented using unary notation. Design a Turing machine that computes the function $f(x,y)$, where $x, y \in 1^+$. $f(x,y) = x - y \; if \; x > y$ $f(x,y) = y - x \; if \; y > x$ $f(x,y) = 0 \; if \; x = y$	08	4	3
9	a	Define Post correspondence problem (PCP) and solve PCP given below: List A: xi = 11, 100, 111; List B: yi = 111, 001, 11 List A: xi = 110, 0011, 0110; List B: yi = 110110, 00, 110	08	3	3
	b	Define Linear Bounded Automata. Design a linear bounded automate to accept the language $L = \{WCW^R \mid W \in (a,b)^*\}$. Give ID for the string $aabcbaa$.	08	4	4
		OR			
10	a	If L_1 and L_2 are recursively enumerable language over Σ, then prove that $L_1 \cap L_2$ and $L_1 \cup L_2$ are recursively enumerable.	06	2	2
	b	Write a note on Chomsky hierarchy.	05	1	1
	c	Define context sensitive grammar and write context sensitive grammar for $L = \{a^n b^m c^n d^m, m, n > 0\}$.	05	3	4